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MAGNETOSTRATIGRAPHY, PALEOMAGNETISM, AND REMANENCE ACQUISITION IN THE TRIASSIC CHUGWATER FORMATION OF WYOMING

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Abstract. The magnetic reversal sequence determined at five sections of the Chugwater Formation can be correlated over a distance of more than 200 km. A detailed study of 203 cores constituting a grid of samples taken from a normal to reverse transition zone at the Red Grade section shows that the transition is characterized by a fairly smooth 180° direction change accompanied by an intensity decrease in about 80 cm of section. The transition occurs in the same way in all vertical columns within the grid and in other columns at the same stratigraphic level up to 80 m from the grid and appears identical to the transition reported by Herrero-Bervera and Helsley at the same lithologic horizon about 1 km from the grid. Thermal and chemical demagnetization of grid samples removes a secondary remanence in the general direction of the earth's present magnetic field and does not reveal the presence of normal and reverse components in the same sample. The transition appears to have been magnetized during a reversal of the earth's field. Samples probably became magnetized in a few hundred years or less while they were close to the deposition surface. Paleomagnetic pole positions for all five sites are very closely clustered, and there appears to be no consistent long-term polar wander from bottom to top in the sections. The mean pole position (46.6°N, 113.5°E; α_{95}=1.88°) is significantly different from pole positions for the Early Triassic Moenkopi Formation from the Colorado Plateau, suggesting that post-Triassic rotation has occurred between the plateau and the basement uplifts of Wyoming.

Introduction

We undertook this magnetic study of the Chugwater Formation in Wyoming for a number of reasons. The initial stimulus was provided by the pioneering work of Picard (1964), who was the first to use magnetic polarity zonation to correlate different sections of sedimentary rocks. Working with natural remanent magnetization (NRM) measurements from the Chugwater Formation in west-central Wyoming, he was successful in establishing the similarity of several major polarity zones at sections as much as 200 km apart. One of our objectives was to determine the extent to which the details of the Chugwater polarity zonation could be improved by the use of a smaller sampling interval and standard demagnetization techniques.

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Any magnetostratigraphic study of red sediments must face the problem of the timing of stable remanence acquisition. The focus of this problem has for the last decade been directed largely toward the Moenkopi Formation of the Colorado Plateau. Papers by Larson et al. (1982) and Purucker et al. (1980) present recent opinions on both sides of this question. It has been one of the more perplexing problems in recent paleomagnetic research that different groups, working on samples that are often from identical sections, find evidence that supports, on the one hand, rapid acqusition of stable remanence and, on the other hand, long-term acquisition. Because Picard's (1964) study indicated that magnetic reversals were reliably recorded by the Chugwater Formation and because of its probable age equivalence with the Moenkopi Formation, we expected that a detailed study of a polarity transition, in addition to the magnetostratigraphic work, might provide further information bearing on the question of remanence acquisition in red beds.

Finally, we felt that it would be useful to collect reliable paleomagnetic data for a substantial portion of the Chugwater Formation. Van der Voo and Grubbs (1977) found a systematic trend of pole position with age for several Chugwater sites and a mean Chugwater pole that is significantly different from that of the Moenkopi Formation (e.g., Helsley and Steiner, 1974). More paleomagnetic data from the Chugwater Formation is needed to evaluate apparent polar wander recorded in Chugwater sediments and the possibility of post-tectonic rotations of Wyoming basement rocks with respect to the Colorado Plateau.

Geologic Setting

The Chugwater Formation crops out at localities throughout most of Wyoming where Laramide uplifts have exposed the Mesozoic section. In the area of our study, south-central to northwestern Wyoming (Figure 1), the Chugwater Formation is exposed in the Rawlins uplift and in structures in and near the Wind River Mountains, an extensive uplift with a Precambrian core. The Chugwater Formation in this area overlies the Dinwoody or Goose Egg formations and is overlain by the Nugget Formation. It is generally subdivided into four units (Love, 1939); from oldest to youngest these are Red Peak Member, Alcova Limestone Member, Crow Mountain Sandstone Member and Popo Agie Member. We sampled within the Red Peak Member and (occasionally) the Alcova Limestone Member.

Lack of diagnostic fossils makes it difficult to establish directly the age of the Chugwater Formation. On the basis of stratigraphic relationships and lithologic correlations, the

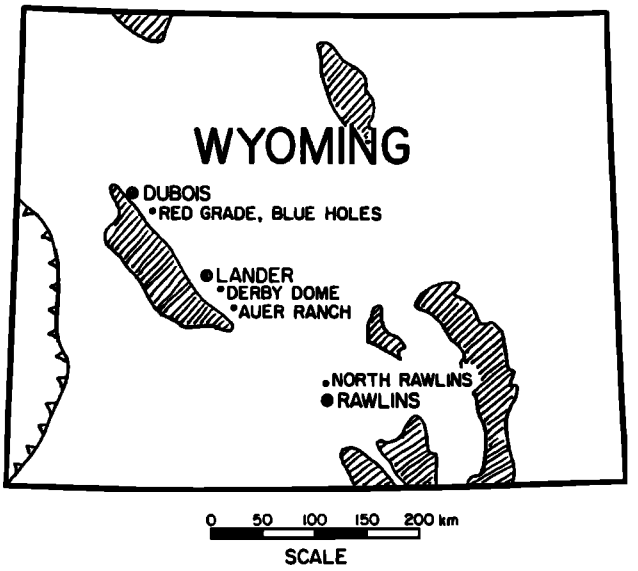


Figure 1. Index map of Wyoming, showing the five sampling locations, the limit of the Western Wyoming thrust belt, and major Precambrian uplifts (hatchured) of Wyoming.

formation is apparently Lower to Upper Triassic (Boyd and Maughan, 1973; High and Picard, 1969), with a possible extensive unconformity between the Crow Mountain Sandstone and Popo Agie Members. The Red Peak and Alcova Limestone members fall entirely within the Lower Triassic. A study of major unconformities in the western interior United States (Pipiringos and O'Sullivan, 1978) suggests that the Red Peak Member is correlative with the Lower Triassic Moenkopi Formation in the Colorado Plateau.

Picard (1964) carried out a paleomagnetic study of the natural remanent magnetization of the Chugwater Formation in west-central Wyoming, including some of the same locations described in this paper. He was able to recognize polarity intervals at four sections and to correlate between sections on this basis. His average sampling interval was quite large (about 10 m), and the lack of demagnetization studies prevented clear identification of polarity in some zones. Nevertheless, the general pattern established by Picard was distinctive enough that Helsley (1969) suggested a correlation of the Moenkopi Formation to the Chugwater Formation, based on the magnetostratigraphic work which had been done on the two formations.

The Chugwater Formation contains thinly bedded to massive units with virtually no indication of channeling and disconformities. It was deposited in a nearshore shallow marine environment or possibly in the intertidal zone along a gentle coastal plain (Picard, 1967). Although there is no direct indication of sedimentation rate, a crude estimate is obtained by dividing the thickness of Lower Triassic rocks (at Red Grade this is about 404 m, which includes 61 m of the Dinwoody Formation) by the length of time allotted to the Lower Triassic, which is about 5 m.y. (Harland et al., 1982). This gives a rate of about 6 cm/10^3 years at the Red Grade section, possibly slightly less to the southeast. Baag and Helsley (1974) used the same approach to estimate a sedimentation rate of about 4 cm/10^3 years for the Moenkopi Formation. Since diastems and erosional unconformities decrease the amount of time available for sedimentation and increase the amount of sedimentation necessary, the average calculated above must be low. An approximate value of 10 cm/10^3 years may be more representative of the rate for the Red Peak Member.

Procedure

The Red Peak Member of the Chugwater Formation was sampled at five separate locations along a NW-SE trend from central to northwestern Wyoming (Figure 1). The Blue Holes and Red Grade sections are only 3 km apart, but other sections are separated by 20-120 km, and the Blue Holes and North Rawlins sections are about 250 km apart. The Alcova Limestone Member, a thin but prominent unit which caps the Red Peak Member, was present at all sections except Blue Holes.

A portable gasoline-powered drill was used to collect a total of 563 oriented cores for magnetostratigraphic studies. The stratigraphic interval between successive cores ranged between 0.6 and 3.0 m. The number of samples and total sampled thickness at each section are as follows: 311 cores from 177 m at Blue Holes (43.43°N, 250.51°E); 102 cores from 95 m at Red Grade (43.42°N, 250.56°E); 22 cores from 32 m at Derby Dome (42.70°N, 251.44°E); 78 cores from 112 m at Auer Ranch (42.60°N, 251.60°E); and 50 cores from 101 m at North Rawlins (41.96°N, 252.66°E).

In addition to the magnetostratigraphic sampling, detailed sampling was done at a normal to reverse polarity transition zone near the bottom of the Red Grade section. This transition zone occurs within a thick siltstone unit, about 1.5 m thick, which is laterally continuous throughout its approximately 600 m exposure at the outcrop. Sampling of the transition took the form of a two-dimensional grid of closely spaced (10 cm) samples and three additional one-dimensional vertical traverses of the same lithologic interval at distances up to 80 m from the grid. A map of the grid and traverse sampling is shown in Figure 2. The grid contains 203 samples; the three traverses contain a total of 38 samples.

Measurements of remanent magnetization were done with a Schonstedt SSM-1 spinner magnetometer and (for most samples) a SCT cryogenic magnetometer. Stepwise thermal demagnetization was performed in a space which was field free to within ±10 nT (10 γ). Some samples were thermally demagnetized in several steps between 200° and 660° C; others were demagnetized only at 630° C. Stepwise chemical demagnetization of 24 specimens (from cores long enough to yield two specimens) was performed by immersion of specimens in cold 8-normal HCl for periods of time up to 384 cumulative hours. Axial holes 6 mm in diameter were drilled in these cores before chemical treatment to increase the surface area exposed to the acid. Thermomagnetic runs were made in air on a recording magnetic balance (Cox and Doell,

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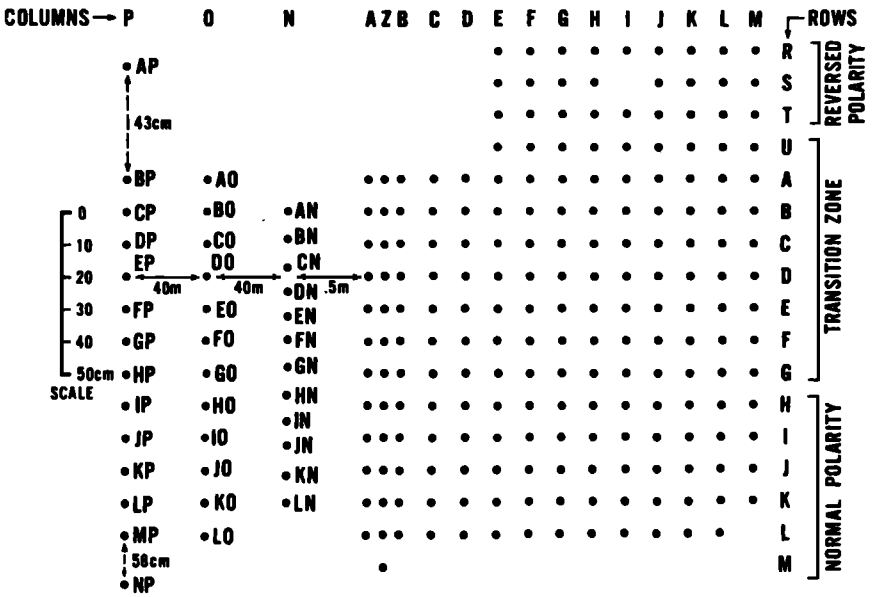


Fig. 2. Location of samples from the Red Grade grid and offset columns. Thin shale partings occur between rows T and U and about 20 cm below row M.

1967). Several polished thin sections were studied by optical microscopy.

Magnetic Mineralogy

A limited study of opaque minerals in the Chugwater Formation was done by Picard (1964). Observations relevant to this study may be summarized as follows: (1) hematite is the dominant opaque mineral, (2) the hematite occurs mostly interstitially and as patches and lenses and also occurs as detrital grains, and (3) magnetite grains are rare; these were generally observed in thin heavy mineral laminations in some samples and are commonly partly altered to hematite. Examination of polished thin sections from the grid at Red Grade gave results consistent with those observations. Pigmentary hematite was ubiquitous. Detrital grains 3-25 μ across were present, but no magnetite grains were observed.

Curie runs on samples with reversed, normal and intermediate polarity from the Red Grade grid were almost identical. They show a dominant contribution from paramagnetic minerals and (to a lesser extent) from hematite. Very slight amounts of magnetite are possible on the basis of (1) a 30% magnetization loss during cooling and (2) a slight kink in the thermal demagnetization curve for the normal sample near 575°C, but none of the curves provides clear evidence for its existence. In any case, behavior of samples during thermal demagnetization also indicates that the magnetite is an insignificant contributor to the remanent magnetization.

These observations indicate that the remanence of the Chugwater Formation resides almost entirely in hematite. The pigmentary hematite is likely secondary and probably carries a chemical remanent magnetization (CRM). Most of the grains of hematite have a detrital shape. Some are magmatic hematite, but many are fragments of martite grains. They may have been martite-hematite or magnetite at the time of deposition and thus may carry either a CRM or depositional remanent magnetization (DRM).

Magnetostratigraphy and Pole Postions

Figure 3 shows the remanence directions at the NRM and 630°C thermal demagnetization levels for the five sections of this study. At all sections, samples had NRM intensities of about 10^{-2} A/m (10^{-5} emu/cc), and these values were reduced by about 20% after demagnetization to 630°C. Samples from the Alcova Limestone, which were collected at Red Grade, Auer Ranch and Derby Dome, were not included in Figure 3 because they could not be effectively demagnetized. We therefore do not know the magnetic polarity or direction of the Alcova Limestone at any site.

The data from the lower 135 m at Blue Holes are markedly "noisier" than data from the other sections. Blue Holes is a natural exposure, but so are the others, except for the lower part of the Red Grade section. Blue Holes is a narrow valley with a series of lakes dammed by travertine deposits. One of the lakes is currently fed by hot springs. Samples from the lower 135 meters were collected from outcrops relatively close to the lakes. We believe that those samples have been heated owing to nearby hydrothermal activity and have acquired greater amounts of secondary remanence than other samples of this study.

Table 1 shows the mean directions before and after demagnetization for normal and reversed sandstone samples, uncorrected for tilt. Demagnetization removes a remanent component that is generally in the present field direction. Thus the normal and reverse directions after demagnetization are more nearly antiparallel, although the dispersion is not significantly less.

The polarity zonation for the five sites, based on remanence data at the 630°C

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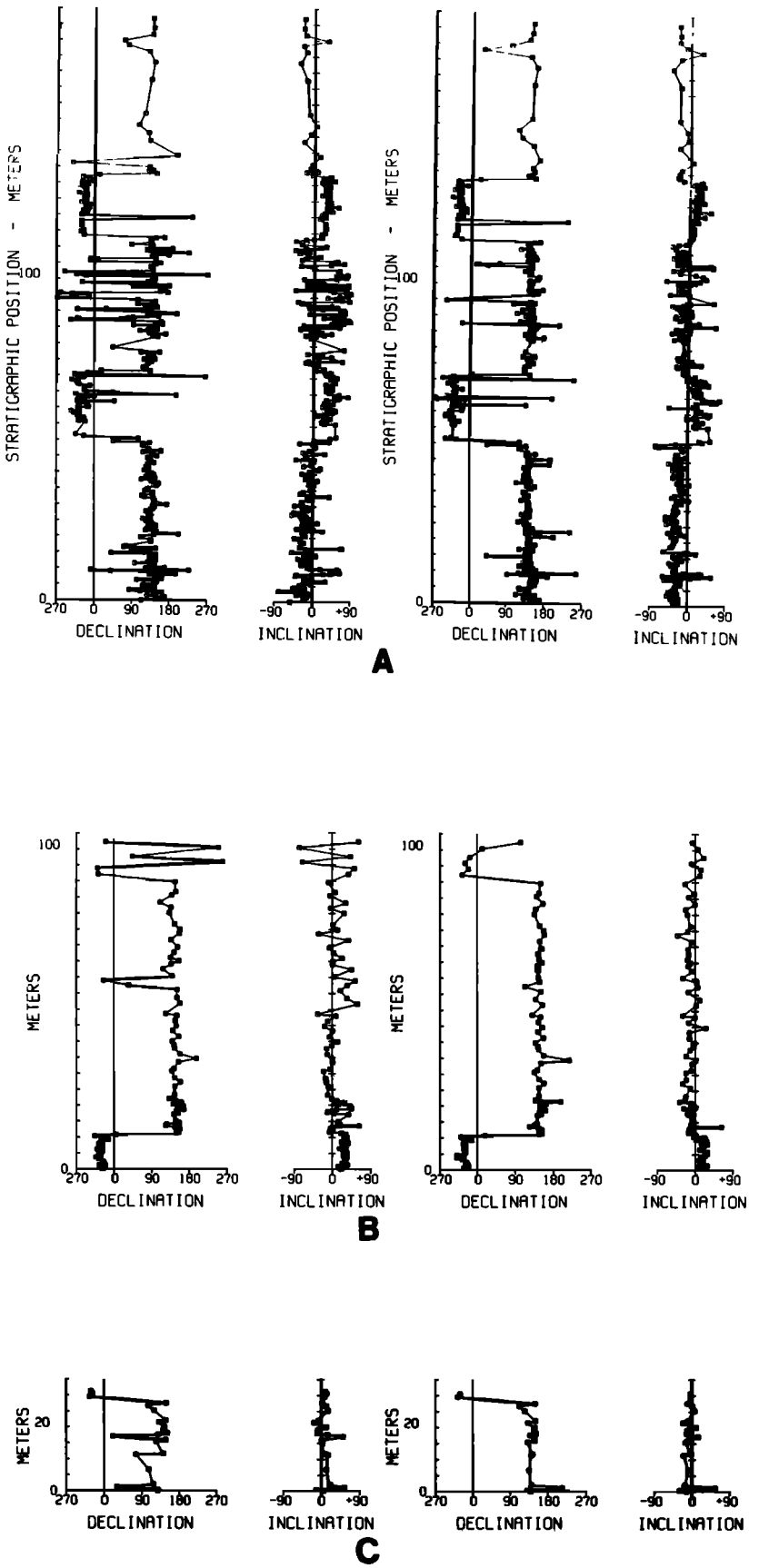


Fig. 3. Declination and inclination of NRM (left) and remanence after 630°C thermal demagnetization (right) before structural correction for a) Blue Holes, b) Red Grade, c) Derby Dome, d) Auer Ranch and e) North Rawlins.

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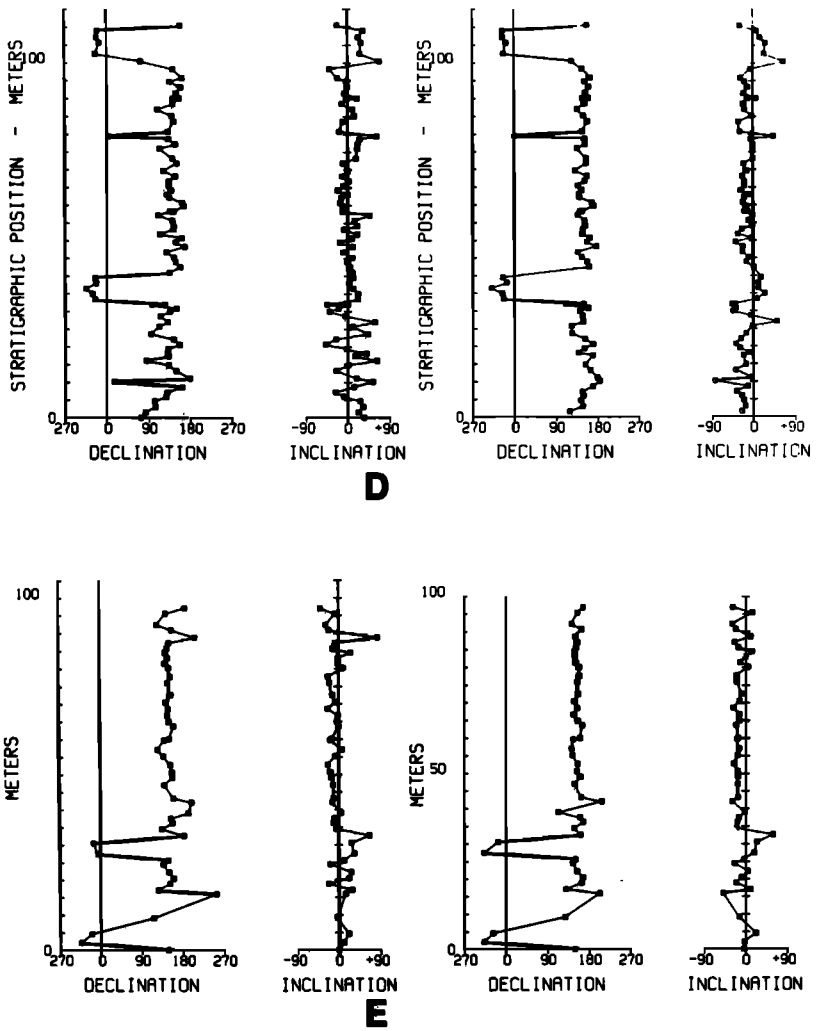


Fig. 3 (continued)

demagnetization level, is shown in Figure 4. Because of the shallow inclination, polarity was assigned mainly on the basis of declination values. For the lowest 135 m at Blue Holes, polarity intervals represented by only one sample are not acceptable because of the thermal problems noted above. For "single-sample" polarity zones at other locations, behavior of the samples during stepwise demagnetization justifies the assignments shown in Figure 4. This method of analysis is always more definitive for reversed samples, because their directions generally move more during demagnetization and because one must always entertain the possibility that a "normal" sample has been remagnetized in the present field. We therefore have more confidence in the short reversed zones at Red Grade, Auer Ranch and North Rawlins than in the short normal zones at Blue Holes and Auer Ranch.

The suggested magnetostratigraphic correlation shown in Figure 4 is an obvious one. All sections are dominantly reversed. Three major normal zones are present which can be correlated at all sections where the appropriate level has been sampled, except at North Rawlins where the upper normal zone is missing. The two lower normal zones are thinner and closer to the Alcova Limestone at the southeast end of the traverse. This is consistent with the progressive thinning of the Red Peak Member in the same direction. The lack of a normal interval below the Alcova Limestone at North Rawlins may be a reflection of a time transgression in its deposition, but erosion and/or lack of deposition are also possible.

Table 2 shows paleomagnetic data for all samples that survived demagnetization at 630°. Normal data were rotated 180°, so mean directions are in the reversed sense; poles are equivalent north magnetic poles. The mean Chugwater pole from this study (46.6°N, 113.5° E, k = 2379.3) is calculated on the basis of the four sites (Blue Holes, Red Grade, Auer Ranch, and North Rawlins) for which the structural corrections are simple and unambiguous. Dips for these four sites ranged between 7° and 16° NE; strike directions between 279° and 342°. Because of the similarity of the tilt corrections and the fact that the stable remanent directions lie close to the tilt axis, the pole positons before correction (mean at 46.3°N, 119.2°E, k = 730.9) moved only a few degrees during correction and became only slightly more clustered. The Derby Dome site, as its name suggests, is on a doubly plunging

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TABLE 1. Mean Directions of Remanence of Samples Before and After 630°C Thermal Demagnetization Before Tilt Correction

	NRM					630°C				
	N	D	I	k	α_{95}	N	D	I	k	α_{95}
Blue Holes										
Normal	77	334.6	36.0	8.0	5.6	72	329.1	17.0	11.1	5.0
Reversed	234	134.1	4.4	3.1	5.2	203	147.6	-13.5	8.7	3.5
Red Grade										
Normal	33	332.5	31.5	34.4	4.2	32	330.2	17.5	51.7	3.4
Reversed	61	147.6	3.8	11.0	5.4	62	149.8	-14.0	21.6	3.8
Derby Dome										
Normal	3	335.0	11.8	-	-	3	326.2	2.1	-	-
Reversed	19	133.7	-10.6	19.3	8.6	19	145.7	-13.3	11.8	10.2
Auer Ranch										
Normal	9	331.1	28.3	31.1	9.3	9	331.4	20.6	31.0	9.4
Reversed	63	139.0	-2.5	8.2	6.6	66	146.0	-22.5	12.9	5.1
North Rawlins										
Normal	5	340.9	23.5	22.4	16.5	4	330.3	20.7	15.2	18.4
Reversed	42	150.7	-6.4	9.0	7.8	46	150.6	-17.0	6.4	9.0

NRM is natural remanent magnetization; N is the number of samples; D is declination; I is inclination; k is the precision parameter (Fisher, 1953); and α_{95} is the radius of the cone around the mean direction drawn at the 95% confidence level. N may differ between the NRM to 630°C levels because samples with intermediate directions were not included in the analyses and because some samples did not survive demagnetization.

anticline. Our confidence in the plunge correction is low so the pole position for that site is not included in the calculaton of the mean. Pole positions for all five sites are within five great circle degrees of each other. The mean pole should be quite representative of this part of the Triassic. Even with the exclusion of the Derby Dome results the data set includes 526 samples, and each of the four remaining sites includes at least 100 m of sections.

Detailed Reversal Study

The grid of samples shown in Figure 2 was drilled in the normal to reverse transition near the bottom of the Red Grade section. The unit from which samples were taken consists of a uniform red siltstone with very thin shale partings between rows T and U and 20 cm below row M. Direction and intensity data associate rows M-H with the lower normal zone, rows G-U with a zone of direction transition, and rows T-R with the upper reversed zone. The normal and reversed zone directions agree well with the mean directions after thermal demagnetization of samples in the normal interval immediately below the grid (D = 329.4°, I = 18.3°, K = 34.3, N = 28) and the reversed interval immediately above it (D = 149.8°, I = -14.0°, K = 21.6, N = 62).

Figure 5 shows a stereonet plot of directions for samples in column K before and after thermal demagnetization. In both cases the directions progress fairly regularly from normal to reverse through the first and second quadrants of the lower hemisphere. The effect of thermal demagnetizaton is, as noted above for samples from the magnetostratigraphic study, to remove a remanence generally along the present field direction.

The directional behavior exhibited by the column K samples is consistent along strike; that is, the transition takes much the same form in all columns in the grid and also in columns N, O, and P, which are displaced 0.5 to 80 m from the grid. The uniformity of behavior is clearly shown by Figure 6, which presents directional data for all grid rows before and after thermal demagnetization. Thermal demagnetization does not usually improve the clustering of the magnetic directions, but that is not surprising since improvement was not significant for the samples from the magnetostratigraphic study.

If a similarly directed secondary magnetization was removed from all samples during demagnetization, great circles drawn through the NRM and 630°C data points should converge near the direction of that secondary component. When this is done for the data in Figure 6, the great circle directions show fairly good convergence near the axial dipole field directon (Figure 7). Note that this technique would not be useful if only antiparallel Triassic vectors were present because all great circles would be equally likely. Constraint in this case is provided by the presence of transition directions (mainly C-G) which lie on a path significantly different from the normal to reverse great circle that passes through the present field direction.

Behavior of individual samples during demagnetization provides further useful information. Twenty four of the cores from the grid yielded two usable specimens each. All of

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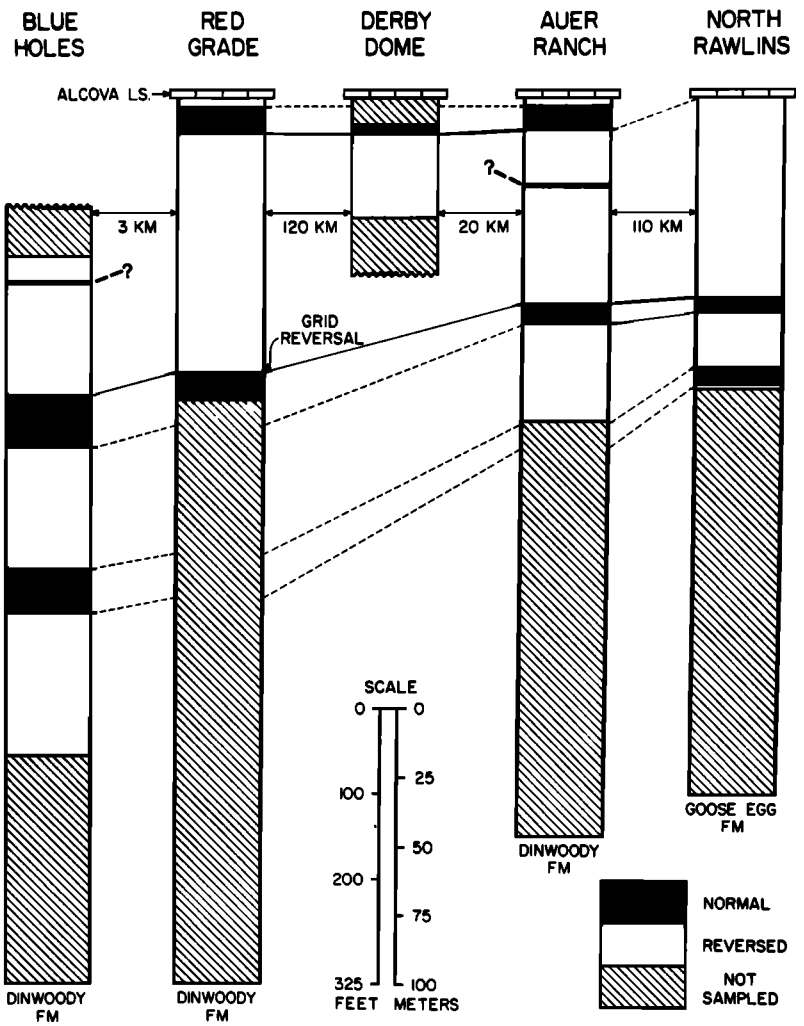


Fig. 4. Proposed magnetostratigraphic correlation of the five sections.

these second specimens were stepwise chemically demagnetized. Of these 24 cores, three were from the reversed zone, seven from the normal zone, and 14 from the transition zone. Figures 8a-8c show the difference vectors (vectors removed during individual chemical demagnetization steps) for the specimens in the three groups. As a comparison, Figures 8d-8f show difference vectors for thermal demagnetization, by group, for the 37 specimens for which stepwise thermal demagnetiztion was performed (not generally from the same samples as the chemically demagnetized specimens).

For samples from the normal and reversed zones, both types of demagnetization remove vectors along the primary direction as expected. In additon, some groups (Figures 8a, 8b and 8c) appear to be losing components in the first and third quadrants, approximately normal to the primary directions. The origin of these components is not clear, as they do not

TABLE 2. Mean Directions and Pole Positions for This Study (After 630°C Demagnetization)

Locality	D	I	N	R	α_{95}	k	East Longitude	North Latitude	Dp	Dm
Blue Holes	150.0	-15.6	305	274.8	2.8	10.1	116.0	45.9	1.4	2.8
Red Grade	151.7	-12.1	96	90.8	3.5	18.2	112.7	45.2	1.8	3.5
Rawlins	154.2	-10.5	48	44.4	5.9	13.2	112.9	47.5	3.0	6.0
Auer Ranch	153.7	-13.9	77	70.6	4.9	12.0	112.4	47.6	2.6	5.0
Derby Dome	152.7	-18.5	22	18.5	13.9	6.0	115.1	49.1	7.5	14.4
Mean--Chugwater Pole	----	-----	4 sites	3.99	1.9	2379.3	113.5	46.6	---	----

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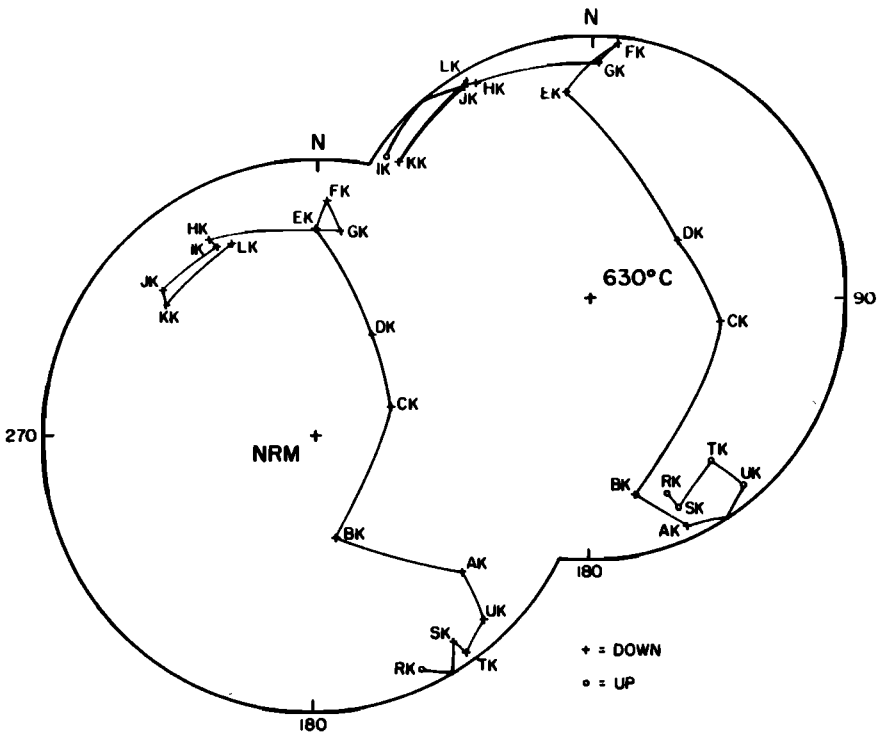


Fig. 5. Equal area projection of remanence directions for samples in grid column K at NRM and 630°C demagnetization levels before structural correction.

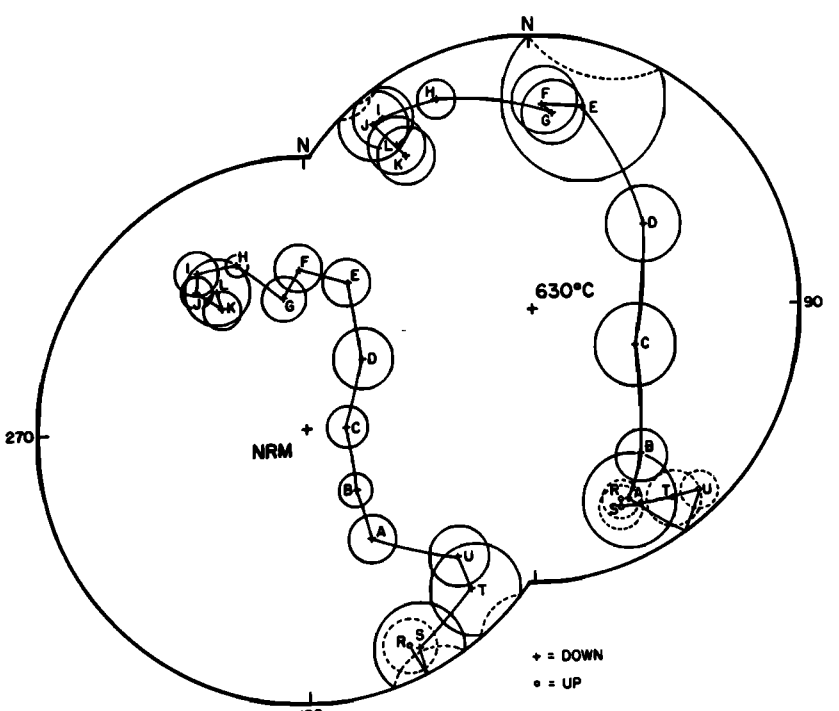


Fig. 6. Equal area projection of mean remanence directions and α_{95} circles for rows from the Red Grade grid at NRM and 630°C demagnetization levels before structural correction.

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represent a known stable field direction for any time between the Triassic and the present. Vectors removed from transition zone samples (Figures 8c and 8f) are by comparison much more randomly directed. There is a preponderance of lower hemisphere components in these two groups, and a slight but noticable clustering of vectors in the thermally demagnetized group (Figure 8f) along the present field direction. Neither of these two groups is dominated by components directed along the primary normal and reversed directions.

Samples from the transition zone lost almost as much remanent intensity during demagnetization as normal and reversed samples (1.67 versus 1.84 x 10^{-3} A/m) but lost a considerably greater percentage of their original remanence (40% versus 27%) as they were less intense to begin with. Figure 9 shows row average intensities and standard deviations for NRM and for remanence after 630°C demagnetizaton. The decrease in remanent intensity is most clearly associated with rows (U-G) that have transitonal directions. Intensities of the normal (H-L) and reversed (R-T) rows are also low in comparison to the mean intensity of 50 samples from the polarity zones imediately above and below the grid (10.0 x 10^{-3} A/m NRM, 8.3 x 10^{-3} A/m after 630°C demagnetization), indicating an intensity decrease of 75-80% in the transition zone.

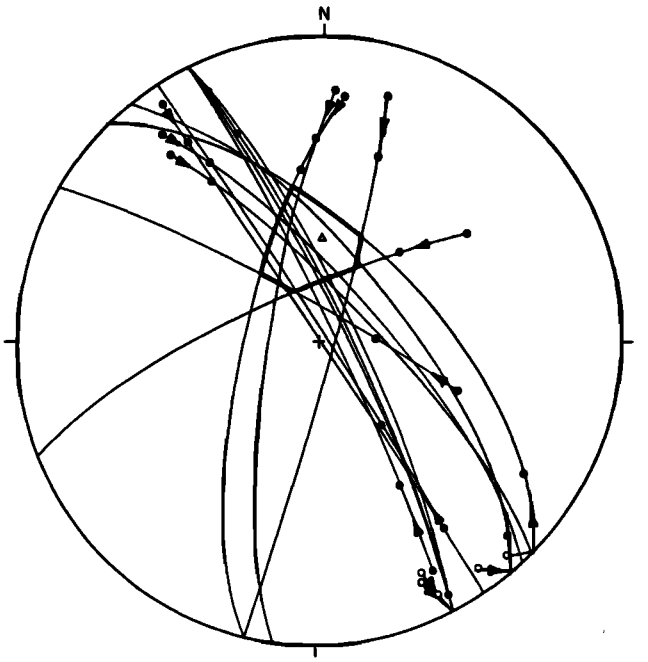


Fig. 7. Great circles extended from NRM directions through 630°C demagnetization directions for rows from the Red Grade grid before structural correction. The heavy line outlines the zone of maximum covergence. The triangle shows the present axial dipole field direction for Red Grade.

Discussion

Magnetostratigraphic Study

Figure 4 shows that most major features of the magnetic polarity zonation in the Red Peak Member of the Chugwater Formation are continuous over the 250-km span of this study. The most notable exception is the lack of a normal zone in the upper part of the North Rawlins section. Thin polarity zones are less consistently observed, as would be expected when the length of a zone is comparable to or less than the sampling interval. In any case, it appears that the Chugwater Formation in central Wyoming contains a remanent magnetization that was acquired in a consistent enough pattern and is well enough preserved to permit correlation on that basis. The probable manner of the acquisition of this remanence is an important question which will be discussed in another section.

Correlation of the Chugwater Formation on the basis of magnetic polarity was first attempted by Picard (1964) over much the same range as this study. Even though we sampled at two of the same sections (Red Grade and North Rawlins), detailed comparison of the two studies is complicated by differences between the studies. Picard, sampling throughout the entire formation, used a much greater sample spacing, while we concentrated on the upper half of the Red Peak Member with a considerably smaller intersample distance; he could easily have missed thin polarity intervals. Furthermore, Picard studied only NRM data, which led him to base some of his correlations on "intermediate normal" and "intermediate reverse" directions. Nevertheless, the conclusions from the two magnetostratigraphic studies are quite similar.

Figure 10 shows a comparison of our magnetostratigraphic data with that of Picard (1964). Our results reproduce the important features of Picard's zonation. The major difference is that we found two normal zones in the North Rawlins section that Picard missed, probably because of his larger sampling interval. Our reversed polarity data just below the Alcova Limestone at Auer Ranch and Red Grade support the idea that Picard's normal interval surrounding the Alcova Limestone is split by a thin reversed zone, as suggested by his "intermediate reversed" directions at South Thermopolis. The solid lines show our suggested correlation within and between the two studies.

Helsley (1969) proposed a correlation between the Moenkopi and Chugwater formations based on rough age equivalence and on the similarity between the magnetic reversal sequence in the lower Moenkopi and Picard's (1964) sequence from the Chugwater. Figure 10 shows the polarity zonation for the entire Moenkopi Formation (Helsley and Steiner, 1974), which includes Helsley's (1969) results. The dotted lines indicate Helsley's (1969) proposed correlation. Affirmation of Picard's basic zonation and discovery of normal polarity just below the Alcova Limestone must be considered as supportive. On the other hand, the overall similarity between the Moenkopi and Chugwater polarity zonations is not compelling. Moenkopi-Chugwater correlation on the basis of magnetostratigraphy must still be regarded as tentative.

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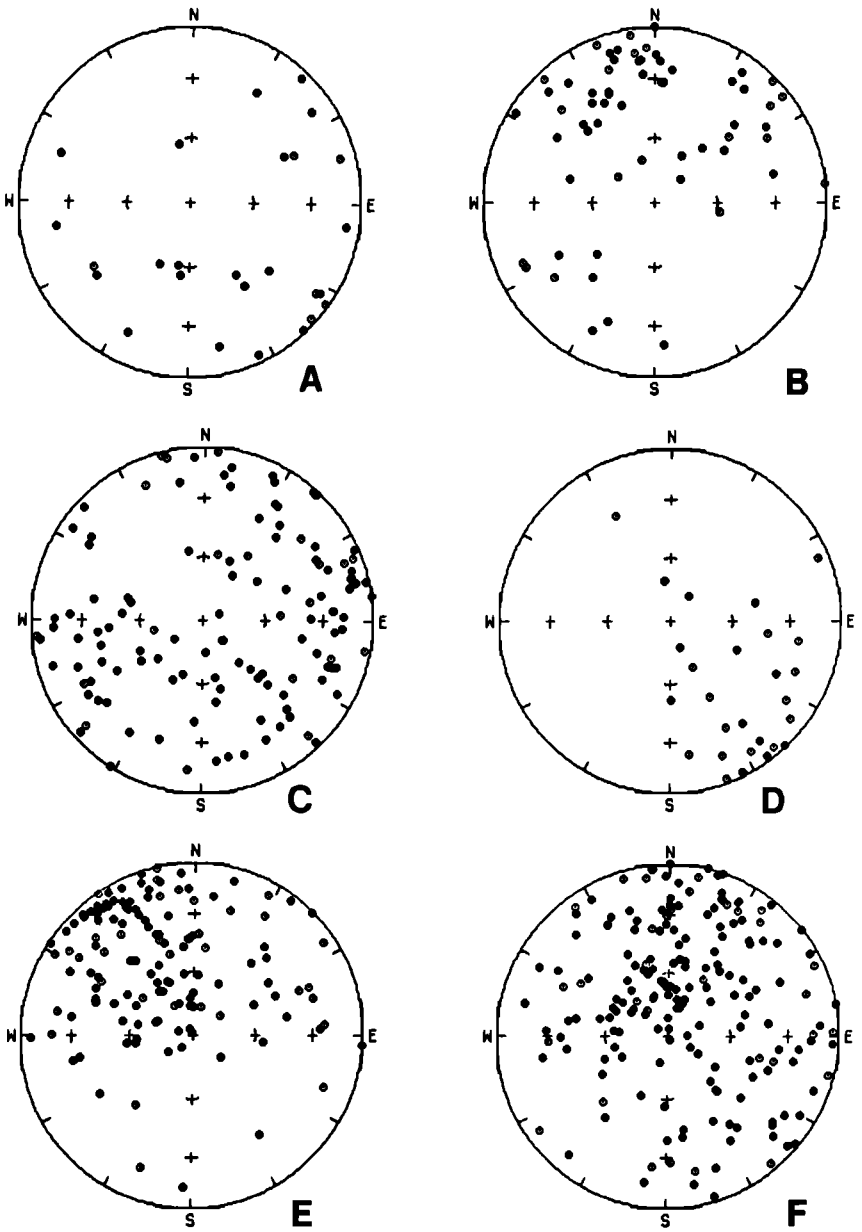


Fig. 8. Vectors removed during stepwise chemical demagnetization for reversed, normal and transition zone samples (a,b,c respectively); vectors removed during stepwise thermal demagnetization for reversed, normal and transition zone samples (d,e,f respectively). All before structural correction.

Pole Positions

Table 3 and Figure 11 compare the paleomagnetic pole position of this study with those from rocks of similar age in western North America. Our pole, from 526 samples, is virtually identical to the Chugwater Formation poles obtained by Van der Voo and Grubbs (1977) for 104 samples from the Red Peak and Popo Agie members near Lander, Wyoming, and by Herrero-Bervera and Helsley (1983) for 25 samples from the Red Peak Member at Red Grade. The Van der Voo and Grubbs (1977) study differs from ours in the sense that their individual site poles showed a systematic clockwise trend with decreasing age while ours are tightly grouped. Seven of their eight total sites are from a single 200-m-thick section of the Red Peak Member, and the seven poles are distributed over a great circle path about 20° in length. This led them to suggest the possibility that the distribution was due to apparent polar wander and that the lower part of the Red Peak Member might be as old as Late Permian. Our magnetostratigraphic data do not show the systematic clockwise rotation of declination with decreasing age that would be consistent with the above observation. Pole postions from all sites are very closely grouped, as noted above. The pole position for the 24 samples from the upper 50 m at North Rawlins (48.5°N, 111.5°E) is deflected clockwise from that for

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105 samples from the lowest 50 m at Blue Holes (48.0°N, 116.1°E), but the difference is only about three great circle degrees.

The same clockwise pole position trend was observed within sections of the lower Moenkopi (Helsley, 1969) and upper Moenkopi (Helsley and Steiner, 1974) formations, although the mean pole for the upper Moenkopi (54.7°N, 104.2° E) is rotated counterclockwise from the lower Moenkopi pole (57.2°N, 98.0°E). Perhaps these trends are caused by incomplete averaging of an anomalous field, a long-term variation of the sort that produced undulatory direction changes in parts of the Moenkopi Formation (Helsley and Steiner, 1974). The existence of similar undulatory behavior in the Chugwater Formation is difficult to establish because our sampling interval averaged much greater than for the Moenkopi studies. The lowest polarity zone at the Blue Holes section, where the sampling density is greater, shows apparent cyclic trends in D and I. Unfortunately, this is also the site most contaminated by secondary remanence.

The mean Chugwater pole position (average of poles 1, 2, and 3; Table 3) at 46.8°N, 113.7°E is significantly different from the mean Moenkopi pole position (average of poles 4, 5, 6, 7) at 56.7°N, 100.9°E. The Early Triassic State Bridge Formation, from the western slope foothills of the Rocky Mountains near the Colorado Plateau, has a pole position (pole 8) that falls between the Chugwater and Moenkopi groups. The Moenkopi and Chugwater poles differ by about 13 great circle degrees, which is clearly significant.

There are three possible explanations for the difference between the Chugwater and Moenkopi pole positions. It could be accounted for by apparent polar wander, if the Chugwater Formation were significantly younger than the Moenkopi Formation. We feel that this explanation is the least likely, given the lack of consistent polar wander in our Chugwater sections and the probable age equivalence of the two formations. The other explanations involve differential motions of the Colorado Plateau with respect to the Chugwater sites to the extent of 12 degrees of rotation and 6 degrees of translation. Paleomagnetic constraints of motion of the Colorado Plateau with respect to stable North America (e.g., Elston and Scott, 1973) are difficult to evaluate because of uncertainties in correlation of rock units within the plateau to units on the stable craton. Finally, we must consider the possibility that Wyoming basement rocks in the study area rotated counterclockwise with respect to stable North America, probably during the Laramide orogeny. The sense and amount of rotation is consistent with models (P. Bird, and others, personal communication, 1983) related to the underthrusting of the North American plate by slabs of the Farallon plate. This sense of rotation is also similar to that of the deflection of the Wyoming basement uplifts relative to the north-south trend of the rest of the Rocky Mountain ranges. The required accompanying northward motion is more difficult to explain. Unfortunately, there are no reliable Early Triassic poles from cratonic North America that could be used to discriminate between these possibilities. Further work will be necessary to explain the difference between the Chugwater poles and other Early Triassic pole positions.

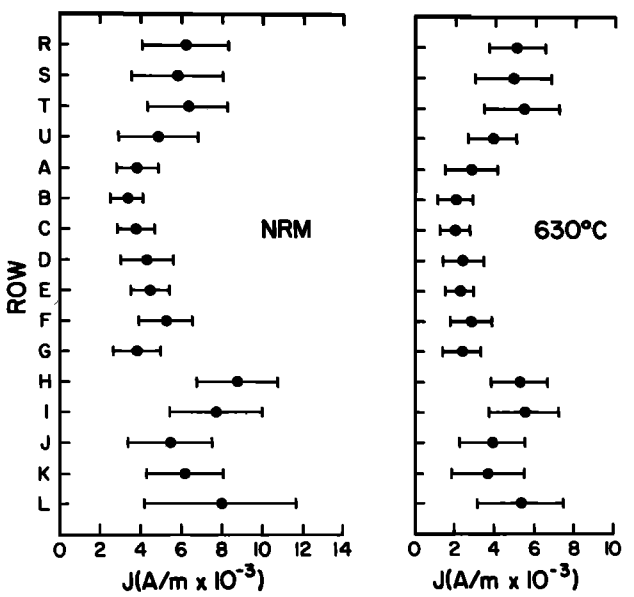


Fig. 9. Mean intensities of remanence for rows from the grid at NRM and 630°C levels. Error bars indicate standard deviations.

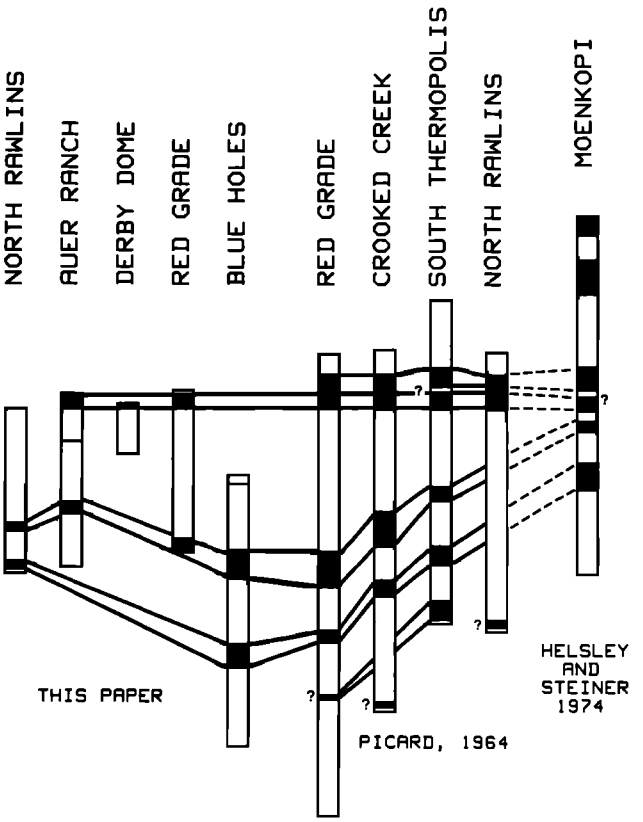


Fig. 10. Magnetic polarity zonation of the Chugwater Formation and suggested correlation to the Moenkopi Formation zonation. Question marks denote intervals assigned on the basis of intermediate NRM directions or anomalous zones.

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TABLE 3. Pole Positions From Early Triassic Rocks of North America

	Formation	Reference	East Longitude	North Latitude	α_{95}	dp	dm	k
1.	Chugwater	this study	113.5	46.6	1.88			2379.3
2.	Chugwater	Herrero-Bervera and Helsley (1983)	115.3	45.4	4.1			47.0
3.	Chugwater	Van der Voo and Grubbs (1977)	112.2	48.5	5.9			86.0
4.	Moenkopi	Helsley (1969) (partial damage)	98.0	57.2		2.5	4.9	84.6
5.	Moenkopi	Helsley and Steiner (1974)(polarity zones)	104.2	54.7		3.0	5.9	110.7
6.	Moenkopi	Baag and Helsley (1974)(average calculated by this study)	100.3	57.0	5.3			547.0
7.	Moenkopi	Purucker et al. (1980)	101.0	58.0		1.3	2.5	10.0
8.	State Bridge	Christensen (1974)	105.7	51.6		1.9	3.8	111.7

Only poles calculated from demagnetized data are listed. Values of k associated with α_{95} values are calculated from site poles; those associated with dp and dm are calculated from sample populations.

Detailed Reversal Study

An important aspect of the detailed reversal study described in this paper involves information we may deduce about the way in which magnetization is acquired and altered in red sandstones. The timing and mechanism of magnetization acquisition has been debated for several years (Elston and Purucker, 1979; Purucker et al., 1980; Walker et al., 1981; Larson et al., 1982). The timing of magnetization is a more important question than the mechanism; red beds become respectable candidates for paleomagnetic study if it can be shown that their magnetization is acquired early, regardless of whether the remanence is a DRM or an early CRM. By "early" in this context we refer to a length of time after depositon which is short compared to the duration of a typical polarity interval for the Triassic, something of the order of a million years or less.

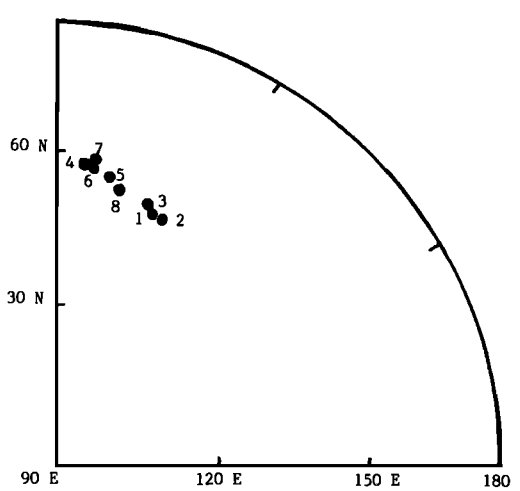


Fig. 11. Paleomagnetic poles for Early Triassic rocks from Western North America. Numbers refer to poles listed in Table 3.

In this study we are dealing with indirect evidence. We have no fold test or conglomerate test data that might allow us to fix the timing of magnetization with regard to a specific deformation event. Rather, we are testing the character of the magnetic information against different possible magnetization histories. Hopefully, we can constrain the reasonable possibilities to a significant degree.

The first critical question involves the basic character of the reversal. Was the transition recorded at Red Grade at the time of a magnetic field reversal? Or does the transition reflect remagnetization of the Red Grade section to a certain level at some much later time during the Triassic?

Remagnetizaton at a much later time appears unlikely for a number of reasons. First of all, remagnetization would have to have varied from partial to complete in almost exactly the same way along strike within the grid, and also 80 m distant at the same lithologic horizon. An independent extension of this condition is provided by the recent study of Herrero-Bervera and Helsley (1983) who sampled the same transiton in three separate traverses about 1 km southwest of our grid. All three of their traverses show that the transition directions are completed in about 80 cm of siltstone section just below a thin clay marker horizon and pass fairly smoothly through the lower first and second quadrants. Remanent intensities drop to 10-25% of the values associated with normal and reversed intervals, and the intensity low occurs over a considerably greater interval than the direction transition. It is clear, then, that the transitional details are recorded in an almost identical manner at the same lithologic

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horizon over horizontal distances of the order of at least 1 km.

We have not studied the details of the polarity transitions at Auer Ranch and North Rawlins. However, if our correlations within the Chugwater Formation are correct, remagnetization to approximately the same level would be required at about the same time over an area more than 200 km across. If the correlation to the Moenkopi reversal sequence is correct, remagnetization is clearly ruled out.

The detailed character of the transition at the Red Grade grid provides more evidence against remagnetization. Since both polarities represent Triassic directions, remagnetizaton during the Triassic would be required. Transition directions would then contain two polarities, and the transition path would be either (1) a shortest great circle between two slightly antiparallel Triassic directions (assuming that the present field overprint could be completely removed), or (2) a great circle through the present field direction (assuming incomplete demagnetization).

The transition path clearly does not pass through the present field direction, which eliminates possibility 2 above. Could transition directions represent mixtures of polarities biased toward the northeast, as in possibility 1? We can test this explanation by examining the average directions of the normal and reversed rows in the grid and also the directions of the normal and reversed intervals above and below the grid.

The mean directions of the polarity intervals immediately below and above the grid are close to antiparallel ($D = 329.4^\circ$, $I = 18.3^\circ$ $D = 149.8^\circ$; $I = 14.0^\circ$). The shortest great circle connecting these directions lies close to the vertical plane in the lower hemisphere. The normally magnetized grid rows (I, J, K, L) have a mean direction with $D = 321.7^\circ$, $I = 10.1^\circ$; the reversed rows (R, S, T) have a mean direction with $D = 152.2^\circ$, $I = -11.6^\circ$. The shortest great circle connecting these directions passes through the southwest quadrant of the upper hemisphere. There is therefore no bias toward the northeast of the directions associated with the normal and reverse zones, as would be required to produce the observed transition path by remagnetization. A third component directed toward the northeast could be invoked to produce the proper bias in the transition directions; if so, it somehow failed to be recorded by the normal and reversed zones above and below.

Finally, we must consider the behavior of individual samples during demagnetization. If transition directions consisted of almost evenly balanced but oppositely directed magnetizations, demagnetization would be expected to disturb the balance and cause the directions to swing toward one or the other polarity and difference vectors should be along the normal and reverse directions. No such tendency is apparent in Figure 8 for either the thermally or chemically demagnetized samples. We did observe that direction changes during chemical demagnetization were generally larger and more erratic than during thermal demagnetization. Chemically demagnetized specimens were always the ones nearest the weathered surface of the outcrop and probably acquired more secondary remanence. In any case, direction swings seldom fall along a normal to reversed great circle.

Thus the weight of the evidence argues against the transition resulting from remagnetization at a much later time. We should now consider whether the observations support the hypothesis that the reversal was recorded while the earth's field was reversing. Obviously some of the observations noted above that were inconsistent with the remagnetization hypothesis may be used in support. These include the consistency of the transition details along strike at and near the Red Grade grid and the Herrero-Bervera and Helsley (1983) traverses and the apparent correlation of that (and other reversals) in the Chugwater Formation over considerable distance.

Records of the earth's magnetic field during a reversal (Goldstein et al., 1969; Watkins, 1969; Dunn et al., 1971; Opdyke et al., 1973; Baag and Helsley, 1974; Hillhouse and Cox, 1976) show that the field experiences a large intensity decrease accompanied by a direction change from one polarity to the other. The intensity decrease usually lasts considerably longer than the time for direction transition, which appears to take a few thousand years. Although the direction change can consist of a series of irregular steps, samples which represent a time average of the field often show a fairly smooth transition. This is consistent with a model in which the dipole field collapses and grows in the opposite direction in the presence of a smaller nondipole field.

The Red Grade grid records a fairly smooth direction change accompanied by an intensity decrease. The zone of intensity decrease is considerably broader than the zone of direction transition (at least twice as broad). Transition intensities after thermal demagnetization drop to 20-25% of the demagnetized intensities in the polarity intervals immediately above and below the grid. The important features of a magnetic reversal are as accurately recorded in the Red Grade grid as could reasonably be expected.

Thus we argue that the samples within the grid acquired their primary magnetizaton while the field was reversing. Since the grid as a whole is resolving an event which lasted a few thousand years, individual samples must have acquired their primary magnetization in a few hundred years or less.

How far below the deposition surface did this happen? Was the magnetization acquired at or soon after deposition, or did the acquisition occur rapidly but long after deposition deep below the surface? We cannot tell from the evidence presented here. However, we are most uncomfortable with a scenario in which magnetization is acquired a million years or so after deposition, which would imply a depth of perhaps a hundred meters or more.

The difficulty with rapid magnetization at depth is that the magnetization must be acquired sequentially in time from bottom to top in a zone whose thickness is very small compared with the depth of burial. For example, we have shown that the magnetization of row D samples was acquired shortly (a few hundred years at most)

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before the magnetization of row C samples. These samples are only 10 cm apart vertically. A mechanism that can (1) magnetize a sediment in a few hundred years at a depth of 100 meters, (2) refuse to remagnetize sediment a few centimeters deeper, and (3) put off magnetizing sediment a few centimeters shallower until a few hundred years later, is most difficult to imagine. Such a mechanism needs to have, at the very least, unusual sensitivity to depth.

We feel that it is far more likely that the magnetization was acquired at or near the deposition surface. The magnetization could be either a DRM or a CRM, but the acquisition mechanism must have operated in such a way as to fix the magnetization in a few hundred years or less. If this is the case, the 80 cm thick transition zone must have been deposited during the few thousand years it took for the field direction to reverse. This implies that the massive siltstone unit recording the grid reversal was deposited at a rate of about 10-50 cm/10^3 years, which is not unreasonable considering the uncertainty in determinations of sedimentation rates and the duration of magnetic reversals.

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